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Demountable Timber Joints for Timber Construction Systems

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Demountable Timber Joints for Timber Construction Systems

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In timber craft development, some important knowledge on demountable timber joints had been lost due to several reasons. The purpose of this research is to evaluate and transform accumulated knowledge on timber joints for contemporary practice and design issues in the case of demountable timber structures. To do so, traditional joints are reviewed and structural, carpentry and construction design concepts are classified. A set of criteria and application fields are indicated for contemporary demountable joints. Examples of demountable joints are depicted with their functional explanations and practical applications.

Introduction

Most contemporary structural timber joints are the products of feasible, mass production and economic construction processes. Inserts called 'Modern Fasteners' -also named mechanical fasteners or connectors- like nails, nail gangs, hangers, ties, claw plates, split rings etc. are used in these joints. Before the introduction of these industry products, there were other ways of connecting timber components, which were to cut and to carve holes into the timber. The basic characteristic of these various connections is that they are all interlocking joints. Interlocking is a property of geometry, which provides demountability to the joint. The demountable timber joint can be simply defined as a sturdy connection of two or more structural components that are to be under some kind of load. These various types of timber joints are found within the floor, wall, ceiling and roof of a building as well as at any intersection of these building parts.

The earliest and simplest structures using timber had to be appropriate for transportation in response to the nomadic livelihood they served. Therefore, whatever joints they used were, of necessity, "demountable". The North American nomads' Tepee, the Mongolian and Turkish Yurt from north eastern and Central Asia and Tent from southern Asia are the best known examples of demountable shelters having demountable joints [1]. The first demountable timber joint can be assumed to be the utilization of forked branch to carry another piece of wood. Tying and untying the knot between the wooden branches provides demountability of these joints. This tie –made out of reed, skin or similar material- is the primitive form of a fastener. Therefore, it can be said that the timber joint was, originally, " a demountable timber joint". This demountability remained a notable feature of these joints, as they developed to meet the increasingly sophisticated needs of an emerging sedentary life in later centuries. Lionel Casson traced the origin of such joints in naval construction of the first century BC [2]. A loose wooden tongue and groove joint with a fastening peg was used to connect each row of planking to the other in boat construction (Fig.1).

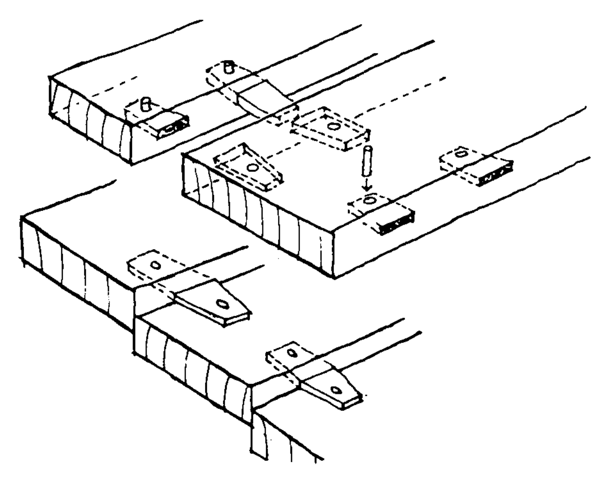

Figure 1: Tongue and groove joints with pegs used in naval construction (Adapted from Casson, 1963) [2].

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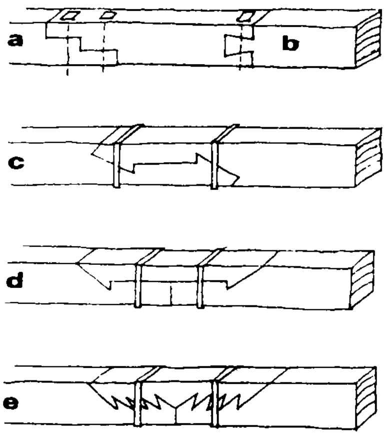

Figure 2: Lengthening joints a) Tabled scarf joint; b) Dovetail joint; c) Compression bending scarf joint; d) Beveled end spliced joint; e) Fingered spliced joint (Adapted from Alberti) [5].

According to Casson, Heredotus describes similar ship joints used in connecting the planks edge and end around 2000 BC in Egypt [3]. Such a configuration was named *harmos*, meaning joint in Greek. Form of a similar root, "harmony" meant "fitting together" in the antiquity [4]. Thus, the word "joint" has the meaning of well fitting. Leon Battista Alberti, in the "Ten Books of Architecture" illustrated examples of interlocking joints with ties for lengthening of timber element (Fig.2) [5]. Another interlocking joint example is found at the roof construction of houses in the Hittite era (1660 BC) [6]. The timber roof planks are joined with notches to provide a structural frame and fascia to carry the earth (Fig.3). The same technique was also used in the roof construction of the Lycia tombs in Anatolia (700-300 BC).

One unique example of a building type composed of demountable joints is the storehouse (grain or corn store). Storehouse is a common example which still exists in a wide range of cultures including Europe, The Black Sea region, central and eastern Asia and Japan (Fig.4). Storehouse is a simple shed enclosing a single space, elevated on posts. It is a combination of skeleton and stacked (log cabin) timber construction. Its potential on demountability is very important in the area of the Eastern Black Sea of Anatolia, since it allows to be moved from one place to another e.g. the owner of the storehouse may carry it elsewhere when they sell the land [7].[1] Demountability established a religious meaning and importance in the periodical disassembly and reconstruction of some Japanese shrines. Interlocking joints were developed in Japan since the 1st Century AD [8]. Through two millennia, the craft has reached to its ultimate limits in design of intricate, precise interlocking joints. Thus, whether used or not, demountable property is inherent in the geometry of joints.

Background and Objectives of the Research

Cursory investigations suggest that there is no source on demountable timber joints that a designer can refer to. It is evident that somewhere along the line of timber craft development, some important knowledge on demountability had been dropped off. This deficiency could be attributed to the introvert organization of the guilds that secured timber knowledge from the outsiders. Japanese carpenters even did not show their tools to the outsiders in order to protect the know-how techniques of the joints and other works. On the other hand, the passing of this craft knowledge gained through hands-on experience and by word of mouth to the next generation; finally it was destined to fade out. Sedentary life provided limited use of demountable joints and structures too. It was certain that with the advent of mechanized production the "more and faster" production forced craftsman to abandon the most vital links of the craft. Not only did this mechanized form of production eliminate the demand for traditional timber craft and its joints, but it also started to eliminate the distinctions of local identity and prevented the use of these joints universally. As a result, the demand for

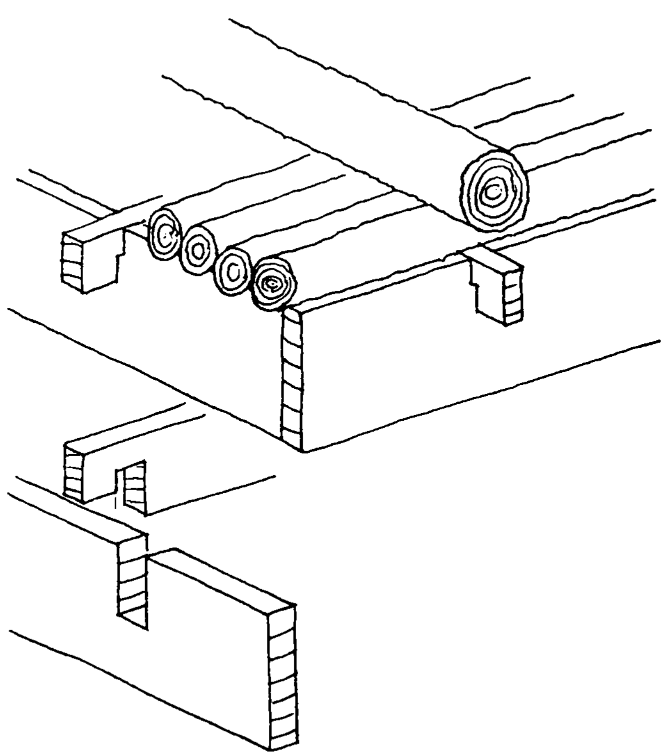

Figure 3: Interlocking notch joint used joining planks on logs of flat roof construction to hold earth behind (Adapted from Naumann, p. 155) [6].

1 This type of storehouse is called "Serender" in Turkish.

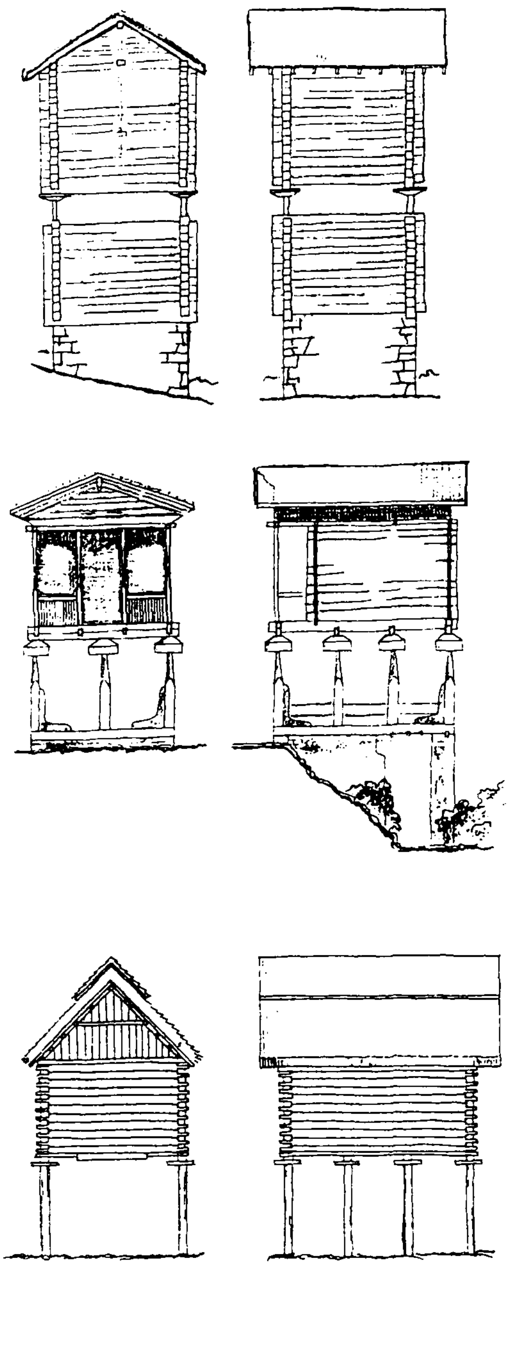

Figure 4: Storehouse examples, Top: Alpine region, Switzerland; Middle: Serander, Turkey; Bottom: Japan.

faster production meant that the teaching of new skills and knowledge became increasingly important. Once this path was taken, other latent factors also came into play so that even further links were lost via the various but rare manuals and handbooks on traditional timber craft. The early manuscripts date back to 14th century. In Europe and Japan the first handbooks of timber craft were printed in the 17th century. Carpenters practiced their craft in the hierarchical nature of the guilds both in Europe and Japan. It nearly took ten years for an apprentice to be a carpenter in the Japanese guilds. They were educated by studying the manuscripts, as well as by watching their masters' work. Design of buildings and joints were prepared by the master carpenter with the use of tools such as, measuring and marking squares, scales, inkpot and pen [9]. Design of a joint was carried on a piece of timber by the use of several types of squares, gauges and long measuring rod. Details and especially joint drawings were the most missing or forgotten items in these early manuscripts. Thus, the traditional knowledge had started to disappear. In Europe, the development of towns during Medieval Age brought the large use of timber in construction. Wars and fires caused the decrease of forests. Then, timber became a scarce building material and the use of timber in buildings was forbidden [10]. Consequently, timber craft and especially joints were forgotten in some places. The various construction systems were so simplified that, it lost its craft techniques in a sense that any carpenter with the barest skill in using an axe and saw could put up a small building anywhere. There was a great effort especially in the 20th century to gain back the knowledge accumulated in 1300 years of Japanese timber culture and the old rules and details were adjusted and refined to modern the Japanese carpentry.

The accuracy of joints became such an ordinary fact in Japanese carpentry that precision was accepted as a minimum requirement in joint making. The geometry of interlocking joints was developed according to available tools and techniques. Moreover, there was a reciprocity between the design of a joint and its cutting process or tools. For instance, in Japanese joint making, starting holes were rarely used but the joint was carved with chisels.[2] Chisels were two types, striking chisels and paring chisels [9]. There were many variations in width, shape, weight, blade slope and size for chisel types. Advanced techniques for Samurai sword forging were also used in chisel manufacturing. Another example for wood cutting tool is the saw. In Japanese tool repertoire, there are different types of saw for cross cut and along grain cut whereas in European tradition one type of saw is used for both purposes. On the other hand, Japanese carpentry did not prefer to use saw unless it was necessary, since the teeth of the saw damages the cellular structure of the wood. Generally Japanese tools were designed for hand and mind skills of the joint maker, but European tools were more robust and practical to use. Technological advances or new designs necessitate new tools and new tools enable more new joint designs. The geometries of traditional interlocking joints are suitable for hand tools; so, may not comply with modern, remote control tools. Therefore, the geometry of these joints must be adapted or redesigned according to contemporary cutting, drilling and shaping techniques. Mortise and tenon, finger and tongue and groove joints are some examples, which are already adapted to modern manufacturing process. From the researcher's observations it becomes evident that serious effort and study are needed: to recover/restore forgotten craft knowledge on demountable timber joints from sources still surviving and to transform those existing timber joints into demountable timber joints. Thus, many intrinsic values of timber joints can be exploited in meeting the persistent demand for contemporary demountable joints.

2 Starting holes are the first holes drilled at corners to mark the hole before carving.

Interlocking Joint Geometry and Demountability

In the review of timber craft, it is observed that demountability was characterized by the following joint systems. Firstly, the interlocking geometry of the joint itself provides demountability, as seen in the joints of storehouses. Secondly, fasteners like rope, skin, etc. that are tied or timber keys and screws provide demountability. Lastly, a combination of the above two items is used. In fact, interlocking joints are those, which are designed to achieve a sturdy connection using the intricately cut form of the joints. Originally, interlocking joints were defined as those joints where stresses are transferred directly through the timber surfaces without the help of fasteners [11]. But in modern times, inserts are also used inside the interlocking joint for reinforcement. Some of the joints that have such interlocking capacity are the bridle, combed, dovetail, finger, forked, gooseneck, housed, half lap, mortise and tenon, scarf, spline or tongue and groove joints. Examples of these joints are depicted in Figures 5 to 10. Ideally, a demountable construction system is one in which the essential load bearing components of a structure are designed to allow them to be taken apart for re-assembly elsewhere and/or at a later time. Today, demountable construction systems may serve a large variety of essential functions. These can include emergency shelters in the aftermath of disasters, as farmers' market shelters, as temporary shelters for recreation, for vacation, sports, entertainment and exhibition activities, as temporary living and work shelters at construction and archaeological sites, as fair and band stands, as spectator seat platforms and as circus tents to cite but a few. These possibilities reveal differences in terms of both the degree of enclosure needed and scale; some may even require elaborate structural solutions, as in the case of sports or entertainment facilities. Also significant is the probable assembly/re-assembly cycle and the resulting wear and tear to which both components and joints will be subjected.

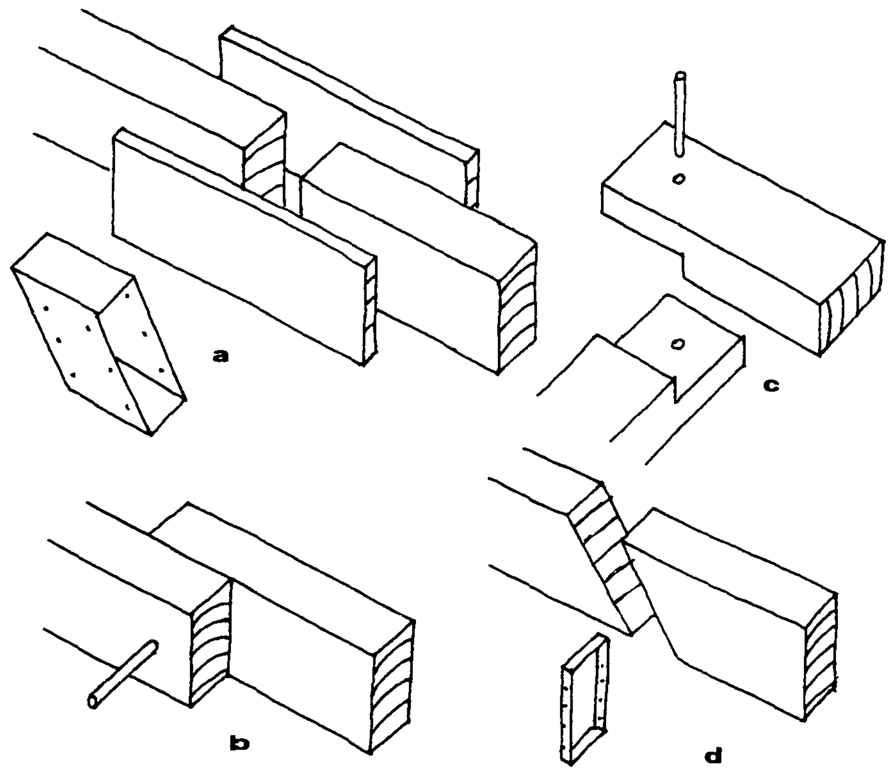

Figure 5: Simple cut joints: a) Butt joint with wooden splice and metal beam seat (left); b) Lap joint with peg; c) Half lap joint with peg; d) Scarf joint with metal tie.

Design Concept of Demountable Joints

Since there is no standard reference regarding demountable joints, the review was subjectively carried out on those timber joints which could be accepted as demountable because of their geometry. In these sources, it was observed that the joints were classified according to the authors' interest. Within this frame, the researcher carried out a survey on timber joint classification and prepared a taxonomy in his previous study [12]. Classifications were primarily based on structural, carpentry and construction concepts, which are essentially addressed three basic questions: "How does the joint resist forces?; "How does the woodworker manufacture the joint?" and "How does the carpenter construct the joint?".

In a timber building, since joints are the weak points, there are a number of structural conditions joints should satisfy. They should be strong enough to transfer working loads to other components and be able to withstand loads such as earthquake and wind. Joints should be located away from critical locations, where large shear forces or bending moments are present. The axis of the main joint components should not shift or change in order to ensure sufficient transfer of forces. Although analyzing and understanding the way a single joint behaves leads to better designs for that joint, joints should not be treated in isolation from rest of the structure, because the structure acts as a single unit. Most modern timber joint designs are based on simple structural analysis, yet there are some cases as in traditional interlocking joints, where ordinary structural analysis may not be reliable because of unpredictable variables and anisotropic behaviour of timber [13].[3] Some important variables are the species of wood, density and strength variation in the same species

3 According to anisotropic character of timber, it behaves different in three axes mainly, parallel, tangential and perpendicular to its grain.

(e.g. early wood and late wood), the moisture content, the effect of chemical preservatives applied, number and location of the knots, long term exposure to unusual stress and strain etc. For this reason, wood engineering codes do not include load transfer information neither for traditional interlocking joints nor for demountable joints [14]. In a demountable timber joint, the most critical factors are the strength of wood, interface friction and withdrawal resistance. Joints must be cut with such precision that the installation tolerances can be minimized to provide maximum interface friction and withdrawal resistance, as it reaches its extreme form such as in Japanese joinery. In joints where inserts are used, a minimum amount of timber should be cut out to avoid weakening the joint.

The method of fabrication and the precision of craft are the most common characteristics to be considered in itemizing carpentry features. Today, joints should be designed to allow easy manufacturing because of economical considerations. The quality of workmanship and the precision in cutting directly effects the stiffness of the joint. Although it was very difficult to achieve this precision in the past, today, it can be obtained by woodworking power tools and computer controlled equipment. The installation, handling, safety, service time, maintenance, labour, storage are some of the primary construction features used in classifying demountable timber joint design. Effective design should allow easy assembly and disassembly without requiring excessive care. The joint design should be such that the joint and its parts are not easily damaged during transportation and storage. Joints should be simple to install and require minimal installation time. Safety in assembly and disassembly, in lifting the components and the manner of fixing it are the other items that should be considered. Repairs and replacement of the joint or its parts, the energy required during construction and a possible need for specialist workers are the other considerations in developing a construction classification. Some of these items are adapted from the Performance Specification Checklist [15] by the researcher [16].

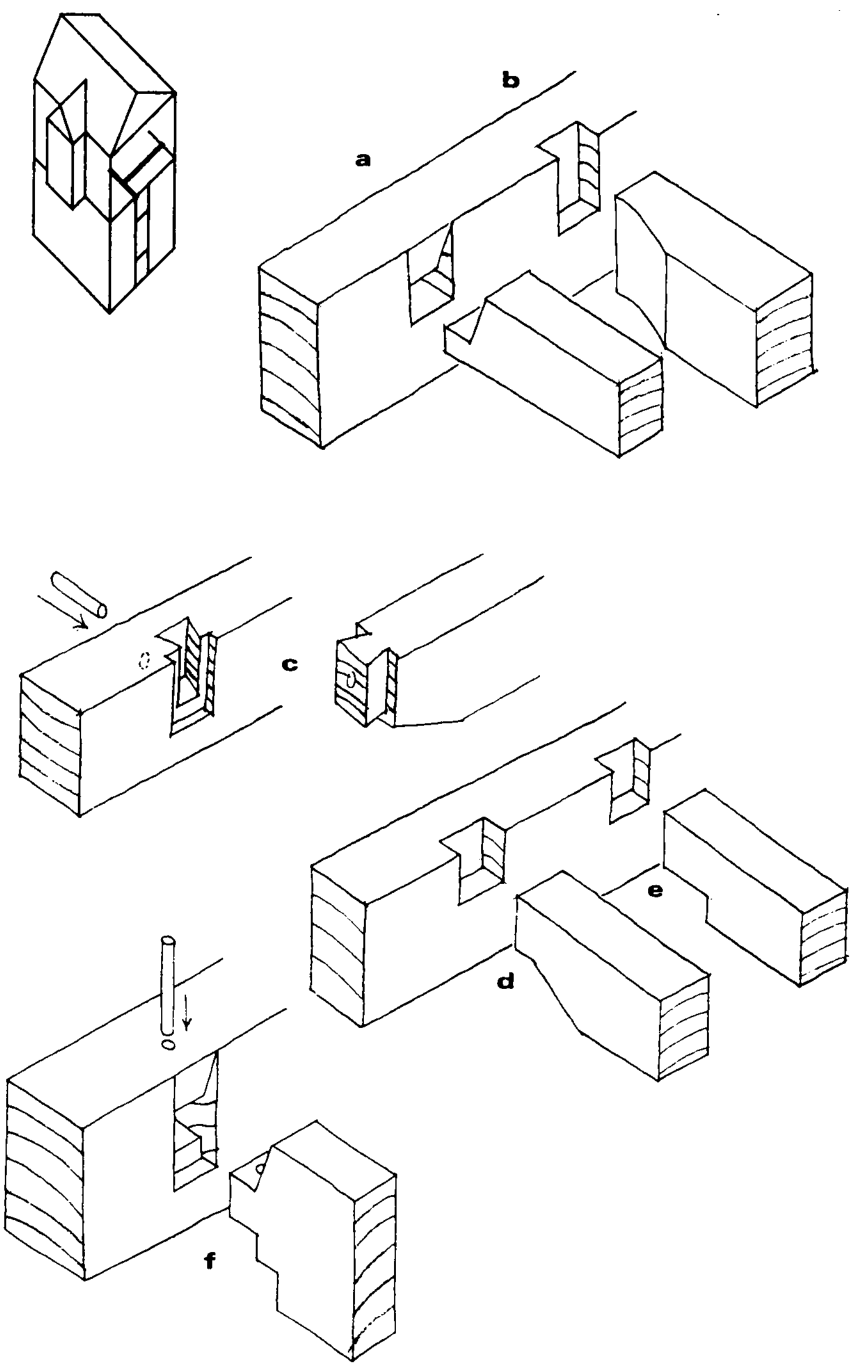

Figure 6: 'T' joints in flooring: a) Beveled horn housed; b) Adzed housed; c) Seated beveled shouldered dovetail; d) Beveled shouldered pocket; e) Notched shoulder pocket; f) Beveled horn seated haunch tenon. The location of the joint in the framing system is shown on the upper left corner of the Figures 6 –10.

Design Concept of Inserts

Structural, carpentry and construction criteria are also valid in classifying inserts, as they are for the joints. Various fasteners or connectors used in joints such as nut and bolt, beam seat, claw plate, screw, split ring, shear plate, tie, hanger, key, pin, wedge, etc. are described by various authors like

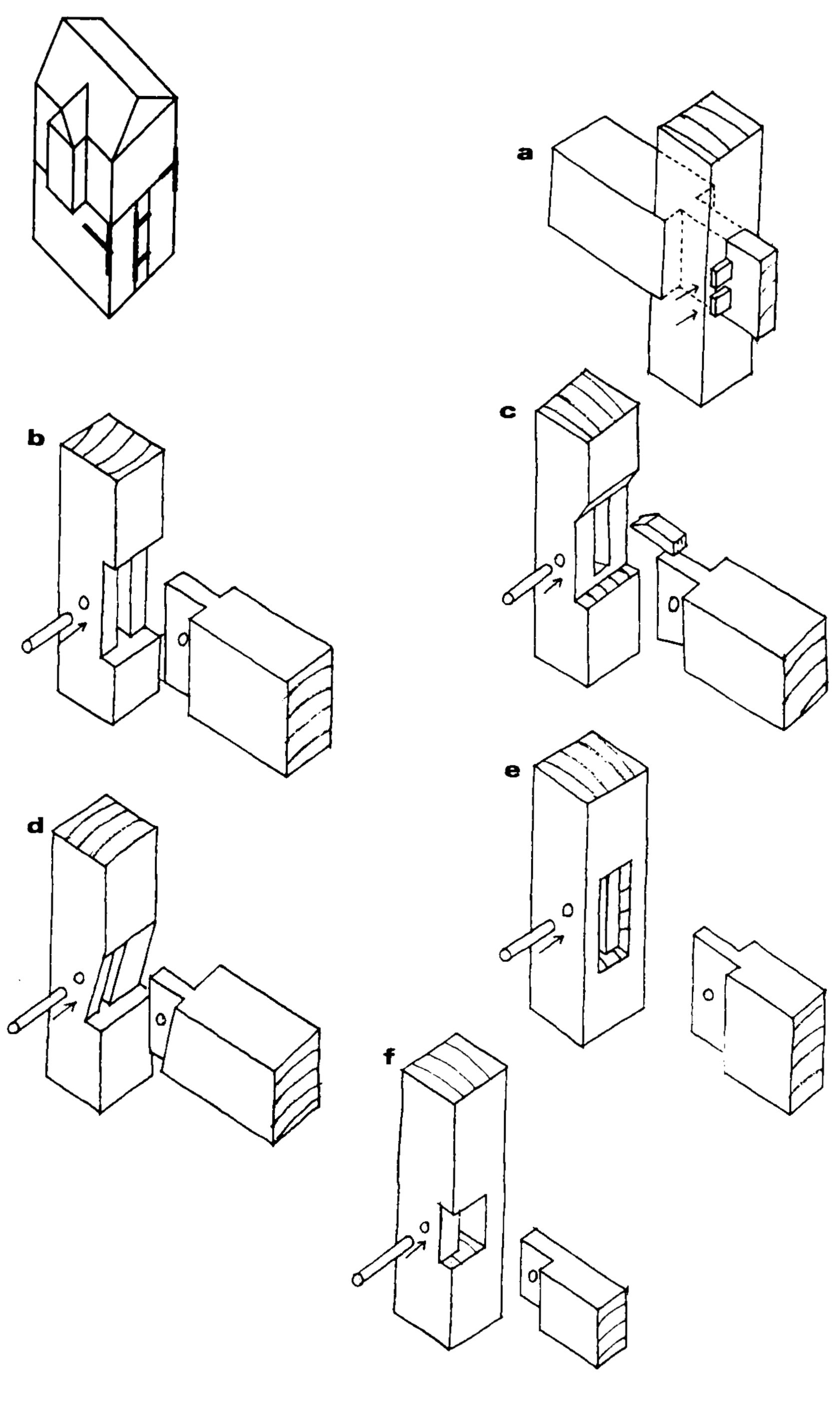

Figure 7: Various 'T' joints: a) Housed extended tenon; b) Seated full mortise and tenon; c) Chased wedged half dovetail; d) Open chased off blind mortise and tenon; e) Seated unequal shouldered mortise and tenon; f) Bareface seated single shouldered tenon. The location of the joint in the framing system is shown on the upper left corner of the Figures 6 –10.

Karlsen [11], Martin [17], Nakahara [18], Oberg [19] and Shaw [13]. In this study, fasteners and connectors are termed "inserts" by the author of this paper [16]. They can be made of wood, metal or plastic. The joints with wooden inserts like the dowel, key, pin or wedge are called "traditional" whereas the others are called "mechanical" or "modern" inserts. Some of these inserts may provide demountability to the joint some do not. In European joints, timber wedges and like used to secure the joint do not provide dismantling. For this reason in Japanese joint craft, these inserts are rarely used; instead, wooden keys are preferred to lock joints. From a structural point of view, inserts used in the joints must be adequately designed for the transfer of forces. Especially inserts are necessary in order to reinforce the joint for more than one type of forces acting in several directions. For instance, an insert in a floor joist should bear to compressive, tensile and shear forces as well as bending moment at the same time. The main considerations in joints designed with inserts are: a) thickness of timber; b) number of inserts; c) type of loading and d) assembly and disassembly. The thickness of the timber component should be adequate to house the insert. Generally inserts bolted or buried in wood provide higher efficiency in strength, ease of fabrication, assembly and even greater fire resistance. In general, no two kinds of inserts have the same deformation characteristics under load. Failure in these bolted joints is sudden whereas in the timber component, it is gradual. The working strength of the inserts is determined empirically, for example, for screws and spikes a reasonable amount of deflection of the hardware is accepted and then the working strength is calculated. Using different kinds of inserts at the same joint is not considered as a good design practice, since one insert can be over loaded before the other reaches its full load carrying capacity. In bolted joints, "joint slip" is a critical factor [20,21].[4] Fewer inserts may be preferred to minimize the labor for installation and potential error. Standard inserts may provide faster assembly. For mass production and mass assembly, templates should be used for dimensioning, cutting, drilling, joining and fastening. If joints are to be cut on site, inserts that are standardized and require less labor should be selected, such as split rings or shear plates. In the choice of inserts, chemicals used in the treatment of timber against fire or biological attack may tend to produce a chemical reaction with metal or plastic. Therefore corrosion resistant materials should be used in inserts.

4 The pre-drilled hole for inserts has a clearance for easy installation. When the joint is loaded, the bolt slips in the hole, thus a deformation occurs that is known as "joint slip".

Contemporary Demountable Structures and Joints

In the evaluation of structural timber joints, the following criteria was proposed by the researcher to consider which timber joint can be accepted as demountable:

1. The joint itself and the inserts should allow dismantling and re-assembly without damaging the main components;
2. The inserts should not suffer damage -mechanical or otherwise- from repeated dis- and re-assemblies, or from being in contact with each other and in re-assembly;
3. The joint should not loose its structural integrity.

In this respect, most traditional interlocking joints can be defined as demountable timber joints. Nailed or glued joints cannot be accepted as demountable timber joints. Other important practical criteria of demountability are as follows:

A. The potential number of assembly and re-assembly cycles of demountable timber joint is dependent on the materials and the design of the joint. Therefore every joint has its own potential number of demountability cycles. In general the number of assembly and disassembly cycles is accepted as follows: the maximum number of disconnection is once a day as for farmers' market shelters and the minimum number of disconnection is once a year or longer in special cases as for vacation shelters.

B. The service life of a demountable timber joint is affected by factors such as the design and materials of the joint, as well as by installation, storage and climatic conditions. The service life of demountable timber joint is a function of following items:

B.1. The complexity/intricacy of the joint: complex, intricate or tiny parts of the joint may allow early break or excessive wear;

B.2. Joint slip: the wear and loss of accuracy of bolted and pre-drilled holes due to repeated re-assembly;

B.3. The excessive stiffening of joints, some inserts like nuts and bolts may cause irreparable defects or deformations on the surface of the joint components;

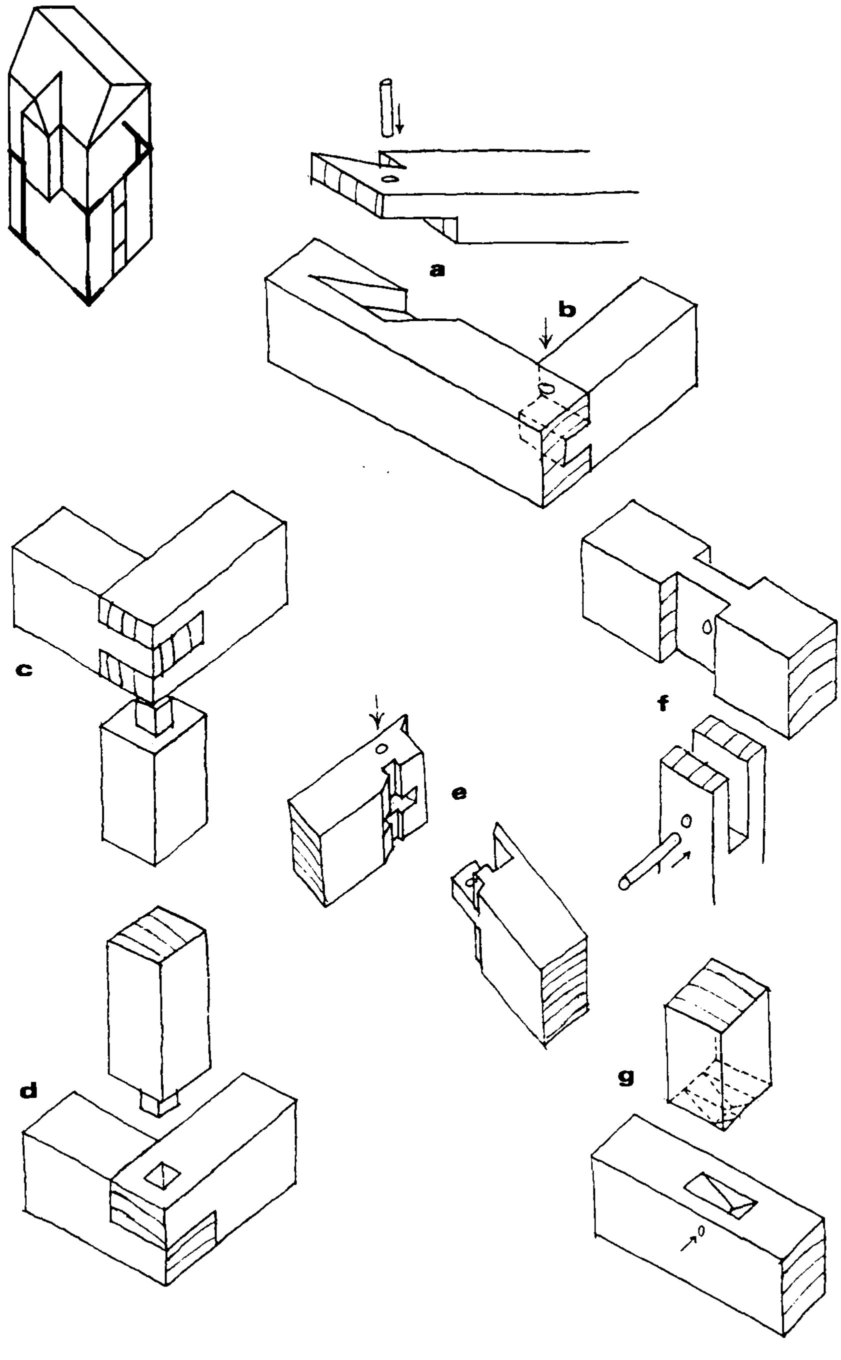

Figure 8: Various joints: a) Half lap halved stop dovetail; b) Half blind mortise and tenon corner; c) Open mortise and tenon with stub tenon post; d) Half lapped corner with stub tenon post; e) Mitered shoulder bladed haunch mortise and tenon; f) Fork tenon; g) Pointed tenon. The location of the joint in the framing system is shown on the upper left corner of the Figures 6 – 10.

B.4. The precision and reasonable tolerances in manufacture and installation;

B.5. Transportation, storage conditions and maintenance;

B.6. Range and/or frequency of change in climatic conditions, especially extremes of heat and humidity;

B.7. Some fasteners used in demountable timber joints can be renewed as required in their service life.

C. The difficulties in assembly and re-assembly caused by the dimensional stability of timber in response to moisture may affect demountability.

D. The structural and construction performance of a demountable timber joint is greatly effected by the stiffness of the joint.

Selected Demountable Joint Examples

The following demountable timber joints are selected from a variety of joints for timber framing construction systems. Sources reviewed include English and translations available in English, originally published in German, Japanese, Norwegian, Russian, Swiss and Turkish. The demountability based on joint geometry, simplicity in installation and manufacturing are the primary criteria in this selection. Joints with simple geometries are classed demountable with the use of inserts and these inserts also generally provide stiffness to the joint (Fig.5). Butt and lap joints have the simplest joint configuration allowing the joint to bear or transfer its load through the contact surface to other components with the help of inserts (Fig.5.a and b). When these joints are loaded parallel to the grain they are not affected by the dimensional change due to temperature and moisture changes. Half lap joints are used at corner joints of wall plates, rafters etc. as depicted in Figure 5.c; whereas, scarf joints are used for lengthening of components (Fig.5.d). Inserts provide resistance especially against shear and some amount against compression or tension.

The joints depicted in Figures 6 to 10 are the examples of joints gaining demountability from interlocking geometry. These examples may also require some kind of inserts (pegs are shown) as illustrated. The 'T' joints are used in horizontal components of floor framing as shown in Figure 6. The housed, pocket and seated joints are used at the connections of horizontal components, e.g. joists with beams, joists with plates, etc. In these joints, a housing may be carved in the primary component to receive the secondary component. The "beveled horn housed joint" (Fig.6.a) provides less cut off fibers along the height of the house when compared to an "adzed housed joint" (Fig.6.b). The "seated beveled shouldered dovetail joint" re-

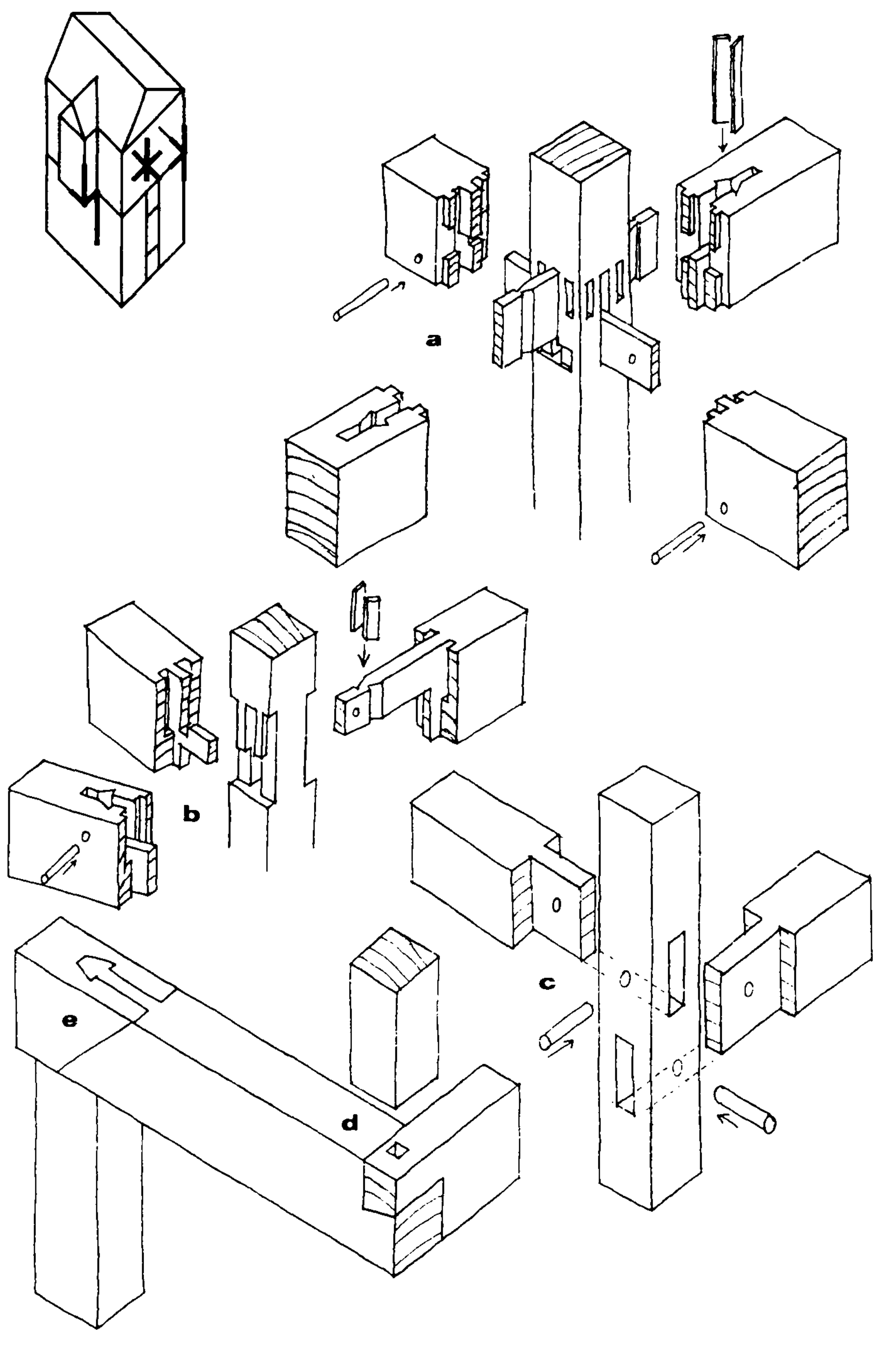

Figure 9: Multiple joints: a) Quadruple ruff keyed spliced mortise and tenon; b) Double ruff keyed haunch mortise and tenon; c) Full mortise and tenon; d) Half lap halved dovetail corner with stub tenon post; e) Under splayed gooseneck lengthening at cantilever with stub tenon post. The location of the joint in the framing system is shown on the upper left corner of the Figures 6 –10.

sists tensile force along the secondary component (Fig.6.c). If the height of the primary component is not sufficient to house the complete cross section of the secondary piece, the depth of the secondary component may be gradually decreased and therefore, pockets are used, as illustrated in the "beveled shouldered pocket" (Fig.6.d) and "notched shouldered pocket joints" (Fig.6.e). The "beveled horn seated haunch tenon joint" is used for heavier loads on a beam and joist connections (Fig.6.f). Some of these joints are also indicated by Brunskill [22], Lloyd [23] and Sobon [24].

The 'T' joints used between vertical components such as in wall framing are shown in Figure 7. In the connection of a vertical component with a horizontal one, e.g. a post with a beam, the mortise and tenon joints are preferred. The "housed extended tenon joint" (Fig.7.a) or a "chased wedged half dovetail joint" (Fig.7.c) is preferred where great tensile force is present. A seat (shallow housing), chase (beveled housing) or housing enables easier installation of the component and provides extra support as illustrated in Figure 7.b, the "seated full mortise and tenon joint". The blind mortise and tenon joints are used at locations under moderate stresses or at locations subject to their own load like the joints of window head and sill components with studs (Fig.7.d). The eccentric location of the tenon may provide easy installation both for the peg and the tenon as shown in Figure 7.d. The "seated unequal shouldered mortise and tenon joint" (Fig.7.e) may also be used in the sill and headers of openings where the width of the component is smaller than the vertical component. The "bareface seated single shouldered tenon" (Fig.7.f) is a simple joint used at the connection of girts. Brunskill [22], Lloyd [23], Sobon [24] and Güngör [25] also explained some of these joints.

The "half lap halved stop dovetail joint" is mostly used in diagonal connections as in floor bracing (Fig.8.a) [26]. The "half blind mortise and tenon joint" is used at corners under light forces (Fig.8.b). The "open mortise and tenon" (Fig.8.c) and "half lap" (Fig.8.d) are simple joints that are suitable at corners of plates with stub tenoned posts. In these corners, open-end grain is protected by using the "mitered shoulder bladed haunch mortise and tenon joint" which obviously requires more labor in manufacturing, as shown in Figure 8.e [8,18]. The "fork tenon joint" is used on the upper end of the post at an intermediate location of the plate (Fig.8.f). The "pointed tenon" (Fig.8.g) at the bottom end of the post provides accurate

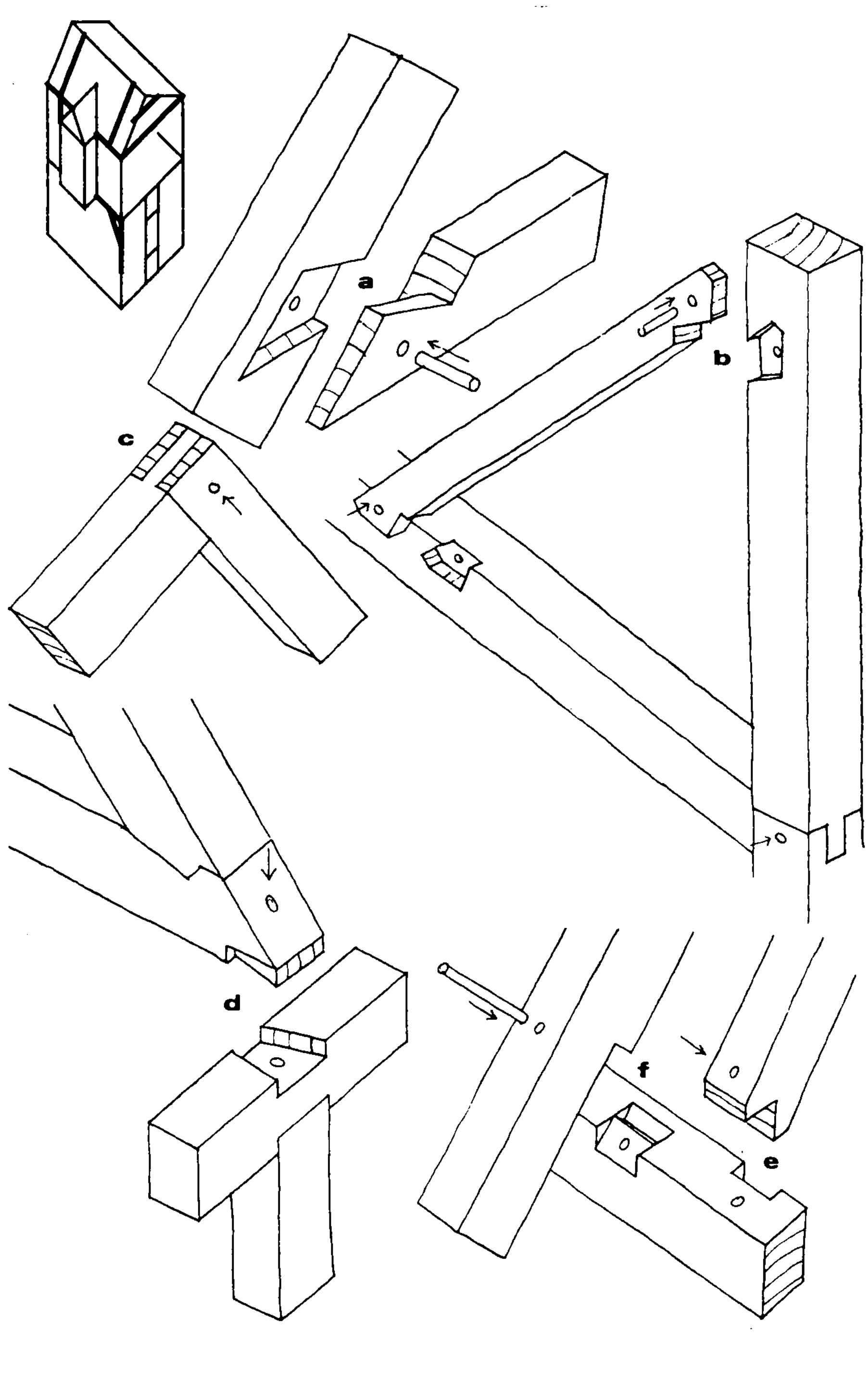

Figure 10: Knee and roofing joints: a) Half lap halved stop dovetail at rafter tie; b) Half lap halved stop dovetail at knee bracing; c) Open mortise and tenon at ridge; d) Front notch rafter with halved dovetail tie beam on housed post and plate; e) Housed bird's mouth; f) Step lapped. The location of the joint in the framing system is shown on the upper left corner of the Figures 6 –10.

installation of the vertical component.

An intricate example of a five component joint is the "quadruple ruff keyed spliced mortise and tenon joint as shown in Figure 9.a. A version of this joint for four components is "double ruff keyed haunch mortise and tenon joint" (Fig.9.b).[5] Both joints are unique examples of Japanese timber craft [18]. Simpler versions of these joints used in the Western World are achieved with full mortise and tenon joints as shown in Figure 9.c [26]. The "half lap halved dovetail joint" (Fig.9.d) is used at connections of cantilever beams with joists. Generally it is advisable to use a single solid beam at cantilevers without a lengthening joint, but the "under splayed gooseneck joint" may be used in certain cases under light loads (Fig.9.e). Some types of joints used in roofing are depicted in Figure 10 [10, 24, 27]. Generally dovetail joints resist tensile forces with the aid of a tapered shape to the collar. The tail gets tighter in the slot as the joint resists tensile force. These joints are efficient for the connections of right angle and diagonal joints (when the grains of two components are perpendicular to each other). Therefore, they are used in rafter ties and knee bracings as illustrated in the "half lap halved stop dovetails" (Fig.10.a) at rafter ties and a "knee bracing joint" (Fig.10.b). When the flare of the tapered surface is small, the dovetail may loosen due to a backward force. The critical factor is the shrinkage of wood in dovetail joints. Since, wood shrinks more tangential to the grain, this effects the tightness of the dovetail joints. An "open mortise and tenon joint" is preferred at the ridge joint of the rafters (Fig.10.c). The "front notch rafter with halved dovetail joint" (Fig.10.d) shown on an inset post and plate is used in roof framing. The "housed bird's mouth joint" is used at connections of a wall plate with a rafter (Fig.10.e). In this joint, the sides of the house prevent lateral movement of the rafter. Many of the bird's mouth joints tend to split along the edge of the notch on the rafter. In rafter roof framing joints, rafters transfer the load of the roof to the wall plates. This load creates thrust; therefore, rafters tend to slide down and must resist uplift forces created by wind and pressure. In the "step lapped joint" (Fig.10.f), the notch resists uplift with the help of inserts.

Conclusion and Discussion

Studying technical and functional aspects of joints cannot fulfill the whole variety of joints. Culture and aesthetic values, way of manufacturing and craft discipline are the other important items that should be considered. Joints were also been the exposition of personal features of the carpenter as in the case of the Japanese master craftsperson. Today, we only see the successful joints of the past. Only those joints, which were suitable for structural, construction, carpentry, climatic and economic considerations survived and the others were forgotten and vanished away.

It is concluded in this research that:

- Demountability is a characteristic of traditional timber joints.
- Demountability is either provided with the interlocking geometry of the joint and/or with the insert.
- Modification of traditional interlocking joint designs to contemporary applications may require reconsideration of the joint geometry according to power tools or computer aided manufacturing techniques.

The review of demountable interlocking joints showed that the Eastern countries -Japan, China- have intricate demountable joints when compared to the Western countries. Especially Japanese timber masters developed very complex interlocking joints requiring excellent craftwork. The reasons for this could be the power of guild organizations on the craft and the authority of social and religious orders of the Japanese culture. Aesthetic considerations became an inseparable part of the joint without putting its primary functions aside. On the other hand, the Western world also developed mechanical fasteners as a substitute for intricate interlocking joints. In this progress, industrialization, mechanization and mass production played the major role. But, this approach limited and even expunged the role of the craft in joint carving. Thus, modern joints with simple geometries like butt and square ends are produced as standard items of mechanized production. The Western industry and its tools also effected Japanese traditional timber craft. After 19^{th} century, the feudal system of the Japanese guilds started to dismantle with the introduction of the Western technology and tools. For instance, the double-edged European saw with some other tools and inserts like steel nails, metal plates, bolts etc. were started to be used in the Japanese joints. Today, we have to utilize the principles of the earlier cultures to contemporary techniques and taste. Modern technology supported the complex traditional timber construction in Japan; whereas in Europe, modern technology created its own way of construction.

Feasibility of demountable joints can be improved with the advanced woodworking tools. The introduction of highly developed, electronic woodworking tools enable the fabrication of intricate interlocking joints. Therefore, interlocking joint manufacturing is not a difficult, manual craft any more. These factory-produced joints may find new application areas in demountable structures like shelters, tents, temporary buildings etc. In a larger perspective, analogies of interlocking timber joint geometry can be applied to other materials like steel, plastics and even to reinforced concrete to provide new joint designs for new demands.

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